Analysis / Polylogarithms

Finite-Difference Determinants

Single-zeta closure, repeated-pole classification, and Stieltjes jets

A research continuation for Vladimir Reshetnikov
and the ProveIt project

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Abstract. We solve a rank-one harmonic-data classification problem raised in the repository's determinantal harmonic-resonance report. For proper rational weights on finitely many translated integer lattices, we give an explicit necessary and sufficient residue test. At nonzero critical decay $x^{-(2r-1)}$, the criterion is particularly simple: rank one is equivalent to being a $(2r-2)$nd finite difference of a proper rational function. Repeated poles and several residue classes are allowed.

This algebraic structure produces an independent-complex-order family of Vandermonde-square sums. At every particle number its diagonal specialization is a finite exponential coefficient plus one Hurwitz zeta value. Polarization allows independent orders and shifts without producing products of zeta functions. Integer orders yield Gamma-weighted complete-harmonic identities. A normally convergent largest-index subtraction gives all Stieltjes jets, with the correct distinction between a hard-cutoff constant and a spectral finite part. The same rank calculation bounds the depth of rational-weight cutoff generators; an explicit repeated-pole example is evaluated in $\mathrm{Li}_{2,2}$ and ordinary polylogarithms.

All infinite-family statements have mathematical proofs. The accompanying programs record 1,037 exact assertions and 29 numerical diagnostics; the latter are not interval certificates. Novelty is assessed against a targeted source audit, not the entire literature. The Gaussian S_6 and revised S_8 conjectures are not resolved.

Repository source anchor: `73f18ef013d5b74eafc883efb2c15210642df12a`

Sources, inspection limits, and proposed integration are in Section 8.

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1 The continuation problem and the precise scope

1.1 What is already available

The canonical manuscript connects nested sums, Gamma identities, Hurwitz zeta jets, and generalized Stieltjes constants. Its preface carefully separates ordinary mathematical proofs, exact finite algebra, and numerical evidence [1]. That distinction is particularly relevant here: a formal rank statement about coefficient matrices does not assert arithmetic independence of the associated special values.

The preserved *Determinantal Harmonic Resonance* report evaluates a Gauss–Vandermonde partition function and treats its first logarithmic resonance. For critical rational weights with simple poles on a common integer grid, its finite moment matrix contains harmonic dependence in only one entry. This gives depth-two cutoff generators even when the number of summation variables is arbitrary [2]. Its further research section asks for a classification with repeated poles, a rank-based depth theorem, and extensions to several residue classes. Those are the specific source questions addressed in this article.

Some nearby questions have already moved beyond that report. The current incoming archive *Polygamma Reconstruction and Resonance* records a solution of the next two-particle resonance and of higher spatial Mellin coefficient questions. The current *Harmonic Tails, Shifts and Twists* archive records the gap-four tail-rank family and further shifted identities [3]. We do not claim those results again. Our principal spectral deformation is a finite difference of an independently variable power; it is not the same deformation as a moving denominator parameter in the earlier Gauss ensemble.

1.2 The three main advances

The algebraic classification in Theorems 6.3 and 6.5 is a necessary and sufficient test, not merely a construction of examples. At nonzero critical decay, it identifies exactly when repeated-pole harmonic data remain rank one. Its finite-difference interpretation explains why arbitrarily high pole multiplicities need not increase the depth of a cutoff generator.

The spectral identity in Theorem 2.3 evaluates a new explicit family for every particle number and every complex order in its convergence domain. The polarized version, Theorem 3.1, permits independent orders and independent shifts. Despite this freedom, only single Hurwitz zeta functions appear: no product of two such functions is needed.

The normal subtraction in Theorem 5.2 supplies ordinary convergent series for every associated Stieltjes jet. Its pole-cancelling factor is explicit. The hard-cutoff constant and spectral finite part are then compared in Corollary 5.4; their difference is not discarded.

1.3 Classical ingredients and novelty qualification

Finite differences, Cauchy–Binet, Gamma differentiation, and the Hurwitz shift formula are classical. Rational summability and discrete residues have an extensive theory, including the work of Chen–Singer and Arreche–Sitaula [6, 7]. We do not present the basic discrete-residue criterion as new. Our use of repeated differences is tied to a prescribed finite Hankel matrix and its critical decay: this gives the explicit harmonic-pencil classification and the special-value identities developed below.

The new statements are research contributions relative to the inspected repository material. The source inspection and literature checks do not establish global historical priority. Nor do they establish that the resulting polylogarithmic representations have minimal arithmetic depth. The

text makes no claim of proof-assistant formalization.

1.4 Conventions

We fix an integer $r \geq 2$ and write

$$d = 2r - 2, \qquad \Delta f(x) = f(x+1) - f(x). \tag{1.1}$$

Thus d is even. A sum over $n_1 < \cdots < n_r$ is ordered; it has no additional factor $r!$. Put

$$\Delta(\boldsymbol{n}) = \prod_{1 \leq i < j \leq r} (n_j - n_i).$$

The repeated use of Δ for a finite-difference operator and a Vandermonde is distinguished by its argument.

Throughout the spectral sections, $\operatorname{Re} a > 0$ and $(n+a)^{-s} = \exp(-s \log(n+a))$ uses the principal logarithm. All parameter derivatives are derivatives of functions on open complex domains, never derivatives inferred from a formula known only at integer orders. We write

$$(s)_k = s(s+1)\cdots(s+k-1), \qquad H_N^{(q)} = \sum_{j=1}^{N} j^{-q}, \qquad H_0^{(q)} = 0.$$

The generalized Stieltjes convention is

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(a) u^m. \tag{1.2}$$

In particular $\gamma_0(a) = -\psi(a)$ and $\gamma_0(1) = \gamma$. A superscript on ζ differentiates its first variable unless an argument derivative is explicitly indicated. We use the standard Hurwitz and Gamma identities with these conventions [8, 9].

A matrix pencil is a matrix whose entries are linear forms in independent formal variables. Its rank means rank over the rational-function field in those variables. This is not the rank of a matrix obtained by treating numerical harmonic values as independent numbers.

2 An all-particle single-zeta determinant identity

2.1 Weights and convergence

Define

$$w_n(s, a) = \frac{\Delta^d (n+a)^{-s}}{d!} = \frac{1}{d!} \sum_{j=0}^{d} (-1)^j \binom{d}{j} (n+a+j)^{-s}. \tag{2.1}$$

For each n, this is entire in s and holomorphic for $\operatorname{Re} a > 0$. Its finite-difference form, rather than a generic Gamma ratio, is the essential feature.

Lemma 2.1 (Decay, positivity, and normal convergence)**.** *Uniformly on compact subsets of* $\mathbb{C} \times \{\operatorname{Re} a > 0\}$*,*

$$w_n(s, a) = c_d(s)(n+a)^{-s-d} + O\big((n+1)^{-\operatorname{Re} s - d - 1}\big), \qquad c_d(s) = \frac{(s)_d}{d!}. \tag{2.2}$$

For $s, a > 0$, $w_n(s,a) > 0$. The partition function

$$\mathcal{Z}_r(s,a) = \sum_{0 \le n_1 < \cdots < n_r} \Delta(\boldsymbol{n})^2 \prod_{i=1}^{r} w_{n_i}(s,a) \tag{2.3}$$

converges absolutely and locally normally for $\operatorname{Re} s > 1$, $\operatorname{Re} a > 0$. *All fixed finite mixed derivatives may be taken termwise there.*

Proof. Repeated use of the fundamental theorem of calculus gives

$$\Delta^d f(x) = \int_{[0,1]^d} f^{(d)}(x + t_1 + \cdots + t_d)\, \mathrm{d}t_1 \cdots \mathrm{d}t_d.$$

Since d is even, $f^{(d)}(x) = (s)_d x^{-s-d}$ for $f(x) = x^{-s}$. Expanding the shifted power to first order proves (2.2), uniformly on the indicated compact sets, including at zeros of $(s)_d$. For $\operatorname{Re} s > 0$ the Laplace representation is

$$w_n(s,a) = \frac{1}{d!\Gamma(s)} \int_0^\infty t^{s-1} e^{-(n+a)t} (1 - e^{-t})^d \, \mathrm{d}t. \tag{2.4}$$

It proves strict positivity for positive real parameters.

For ordered nonnegative indices, $|\Delta(\boldsymbol{n})|^2 \le \prod_i (n_i + 1)^{2(i-1)}$. After dropping the ordering, a majorant for (2.3) is therefore a product of convergent one-dimensional series: its least rapidly decaying factor has exponent $-\operatorname{Re} s$. Local normal convergence follows for $\operatorname{Re} s > 1$. Differentiation is justified on slightly larger compact neighborhoods by the same argument, or by the corresponding logarithmic factors and Cauchy estimates. □

2.2 The finite endpoint matrix

For $0 \le k \le d$, set

$$E_k(s,a) = -\frac{1}{d!} \sum_{j=1}^{d} (-1)^j \binom{d}{j} \sum_{\ell=0}^{j-1} (\ell - j)^k (\ell + a)^{-s}. \tag{2.5}$$

Define

$$E(s,a) = (E_{i+j}(s,a))_{0 \le i,j < r}, \qquad U_r(s,a) = \det E(s,a), \qquad V_r(s,a) = \det(E_{i+j}(s,a))_{0 \le i,j < r-1}. \tag{2.6}$$

These are finite expressions, entire in s. Their coefficients involve only finitely many powers $(a+\ell)^{-s}$; no infinite special-function coordinate is hidden in U_r or V_r.

Lemma 2.2 (Moment cancellation)**.** *For* $\operatorname{Re} s > 1$ *and* $0 \le k \le d$,

$$\mu_k(s,a) := \sum_{n \ge 0} n^k w_n(s,a) = E_k(s,a) + \mathbf{1}_{k=d} \zeta(s,a). \tag{2.7}$$

More generally, for every integer $N \ge 0$ *and every* $s \in \mathbb{C}$,

$$\sum_{n=0}^{N} n^k w_n(s,a) = E_k(s,a) + \mathbf{1}_{k=d} \sum_{n=0}^{N} (n+a)^{-s} + R_{k,N}(s,a), \tag{2.8}$$

$$R_{k,N}(s,a) = \frac{1}{d!} \sum_{j=1}^{d} (-1)^j \binom{d}{j} \sum_{\ell=N+1}^{N+j} (\ell - j)^k (\ell + a)^{-s}. \tag{2.9}$$

Here $0^0 = 1$ *in a zeroth moment.*

Proof. The polynomial finite-difference identities are

$$\sum_{j=0}^{d}(-1)^j\binom{d}{j}(x-j)^k=\begin{cases}0, & k<d,\\ d!, & k=d.\end{cases} \tag{2.10}$$

Insert (2.1), shift $\ell=n+j$, and use

$$\sum_{\ell=j}^{N+j}F(\ell)=\sum_{\ell=0}^{N}F(\ell)-\sum_{\ell=0}^{j-1}F(\ell)+\sum_{\ell=N+1}^{N+j}F(\ell).$$

This finite identity remains valid when the subtracted and added ranges overlap. It proves (2.8) for all s.

For the infinite identity, first take $\operatorname{Re} s>d+1$, so all separately shifted series converge. Equation (2.10) leaves only $\zeta(s,a)$ at $k=d$ and the finite lower boundary (2.5). Both sides of (2.7) are holomorphic on $\operatorname{Re} s>1$ by Lemma 2.1, so the identity theorem proves the full stated domain. This step avoids an illicit splitting into divergent series. □

Theorem 2.3 (Single-zeta closure at every particle number)**.** *For $r\geq 2$, $\operatorname{Re} a>0$, and $\operatorname{Re} s>1$,*

$$\boxed{\mathcal{Z}_r(s,a)=U_r(s,a)+V_r(s,a)\zeta(s,a).} \tag{2.11}$$

Consequently (2.3) has a meromorphic continuation to the whole s-plane with at most a simple pole at $s=1$. Its residue is $V_r(1,a)$. For $\operatorname{Re} a>0$ this residue is nonzero and equals

$$V_r(1,a)=\prod_{j=0}^{r-2}\frac{j!\,\Gamma(a+j)\Gamma(j+1)\Gamma(r-j)}{\Gamma(2r-1-j)\Gamma(a+2r-2-j)}. \tag{2.12}$$

For positive real a, it is strictly positive.

Proof. Finite Cauchy–Binet applied to the monomial evaluation matrix gives

$$\sum_{0\leq n_1<\cdots<n_r\leq N}\Delta(\boldsymbol{n})^2\prod_i w_{n_i}=\det\left(\sum_{n=0}^{N}n^{i+j}w_n\right)_{0\leq i,j<r}.$$

Pass to the limit using normal convergence and Lemma 2.2. The limiting matrix is

$$E(s,a)+\zeta(s,a)\boldsymbol{e}\boldsymbol{e}^{\mathsf{T}},\qquad \boldsymbol{e}=(0,\dots,0,1)^{\mathsf{T}}.$$

The determinant is affine in its bottom-right entry, with coefficient V_r. This proves (2.11). The standard meromorphic continuation of Hurwitz zeta gives the continuation statement [8].

At $s=1$, $w_n(1,a)=\Gamma(n+a)/\Gamma(n+a+d+1)$. The matrix defining V_r is the moment matrix of an $(r-1)$-particle positive ensemble for $a>0$; its series converges, since its highest moment has degree $d-2$. Thus $V_r(1,a)>0$. The product evaluation is a specialization of the Gauss–Vandermonde calculation in the source report [2]; Appendix A gives the finite factorial-determinant derivation with the present normalization. The product has neither zeros nor poles for $\operatorname{Re} a>0$, proving nonvanishing there. □

2.3 An explicit two-particle identity

For $r = 2$, put $A = a^{-s}$ and $B = (a+1)^{-s}$. Directly from (2.5),

$$E_0 = \frac{A-B}{2}, \qquad E_1 = \frac{B}{2}, \qquad E_2 = -A - \frac{B}{2}.$$

Therefore

$$\boxed{\mathcal{Z}_2(s,a) = \frac{a^{-s} - (a+1)^{-s}}{2}\zeta(s,a) - \frac{a^{-2s}}{2} + \frac{[a(a+1)]^{-s}}{4}.} \tag{2.13}$$

This identity holds initially for $\operatorname{Re} s > 1$ and gives the continuation elsewhere. At $a = 1$ and integer $m \geq 2$,

$$\mathcal{Z}_2(m,1) = \frac{1-2^{-m}}{2}\zeta(m) - \frac{1}{2} + 2^{-m-2}. \tag{2.14}$$

For example,

$$\mathcal{Z}_2(2,1) = \frac{\pi^2 - 7}{16}, \qquad \mathcal{Z}_2(3,1) = \frac{14\zeta(3) - 15}{32}.$$

The three-particle specializations include

$$\mathcal{Z}_3(2,1) = \frac{928\zeta(2) - 1325}{5971968}, \tag{2.15}$$

$$\mathcal{Z}_3(3,1) = \frac{333120\zeta(3) - 390083}{2579890176}. \tag{2.16}$$

These are exact determinant evaluations, not fitted numerical relations.

2.4 A specialization anomaly that must not be suppressed

At $s = 0$, every weight in (2.1) vanishes, whereas (2.13) gives

$$\mathcal{Z}_2^{\mathrm{cont}}(0,a) = -\frac{1}{4}. \tag{2.17}$$

Similarly, the pointwise weights at $s = -1$ vanish for $r = 2$, but

$$\mathcal{Z}_2^{\mathrm{cont}}(-1,a) = \frac{1}{24}.$$

There is no contradiction. A series that is identically zero at an isolated parameter need not converge normally in a neighborhood of that parameter. The continuation is defined from the open half-plane $\operatorname{Re} s > 1$. Consequently, statements outside that half-plane refer to this continuation, not automatically to a newly specialized pointwise series. This distinction will also be retained when discussing finite parts and primitives.

3 Independent spectral orders and independent shifts

3.1 A mixed determinant with no mixed zeta products

Let (s_ν, a_ν), $1 \leq \nu \leq r$, be independent pairs with $\operatorname{Re} s_\nu > 1$ and $\operatorname{Re} a_\nu > 0$. The finite-difference order is still $d = 2r - 2$ for every weight. Define the averaged symmetrization

$$\mathcal{Z}_r^{\mathrm{sym}}(\boldsymbol{s},\boldsymbol{a}) = \frac{1}{r!}\sum_{\sigma\in S_r}\sum_{0\leq n_1<\cdots<n_r} \Delta(\boldsymbol{n})^2 \prod_{i=1}^{r} w_{n_i}(s_{\sigma(i)}, a_{\sigma(i)}). \tag{3.1}$$

When all pairs coincide, this is exactly $\mathcal{Z}_r(s,a)$, not $r!\mathcal{Z}_r(s,a)$.

For independent formal variables $t_1,\dots,t_r$, write

$$E(\boldsymbol{t})=\sum_{\nu=1}^{r} t_\nu E(s_\nu,a_\nu), \qquad C(\boldsymbol{t})=\det E(\boldsymbol{t})_{0\le i,j<r-1}.$$

The latter is a homogeneous polynomial of degree $r-1$.

Theorem 3.1 (Polarized single-zeta closure)**.** *The series* (3.1) *is absolutely and locally normally convergent in the stated domain, and*

$$\begin{aligned}\mathcal{Z}_r^{\mathrm{sym}}(\boldsymbol{s},\boldsymbol{a}) &= \frac{1}{r!}[t_1\cdots t_r]\det E(\boldsymbol{t})\\ &\quad+\frac{1}{r!}\sum_{\nu=1}^{r}\zeta(s_\nu,a_\nu)\left[\prod_{\kappa\neq\nu} t_\kappa\right] C(\boldsymbol{t}).\end{aligned} \tag{3.2}$$

The coefficient expressions are finite and entire in all spectral variables. Thus the joint continuation contains only a linear combination of single Hurwitz zeta functions. It has no products of zeta functions and no product poles at intersections of $s_\nu=1$.

Proof. Each term in the finite symmetrization satisfies the product majorant from Lemma 2.1, using $\min_\nu \operatorname{Re} s_\nu>1$ on compact subsets. Apply Cauchy–Binet to the combined weight $\sum_\nu t_\nu w_n(s_\nu,a_\nu)$. Its moment matrix is

$$E(\boldsymbol{t})+\left(\sum_\nu t_\nu\zeta(s_\nu,a_\nu)\right)\boldsymbol{e}\boldsymbol{e}^{\mathsf{T}}.$$

Its determinant is

$$\det E(\boldsymbol{t})+\left(\sum_\nu t_\nu\zeta(s_\nu,a_\nu)\right)C(\boldsymbol{t}).$$

On the series side, the coefficient of $t_1\cdots t_r$ sums all $r!$ assignments of the labeled weights to the ordered particles. Divide by $r!$. This proves the formula and its precise normalization. The meromorphic conclusion follows because the displayed finite coefficients are entire in the independent spectral variables. □

Remark 3.2 (Why independent shifts do not increase the rank). Each separate shift creates its own Hurwitz zeta coordinate, but every coordinate enters the same bottom-right moment entry. The number of special-function coordinates and the rank of the matrix subspace carrying them are different quantities. This is the mechanism behind the multiple-residue-class extension in Section 6.

For $r=2$ and a common shift a, set $A_s=a^{-s}$ and $B_s=(a+1)^{-s}$. Equation (3.2) reads

$$\begin{aligned}\mathcal{Z}_2^{\mathrm{sym}}(s,t;a) &= \frac{A_s-B_s}{4}\zeta(t,a)+\frac{A_t-B_t}{4}\zeta(s,a)\\ &\quad-\frac{A_sA_t}{2}+\frac{A_sB_t+A_tB_s}{8}.\end{aligned} \tag{3.3}$$

In particular,

$$\mathcal{Z}_2^{\mathrm{sym}}(2,3;1)=\frac{7}{32}\zeta(2)+\frac{3}{16}\zeta(3)-\frac{29}{64}. \tag{3.4}$$

The unaveraged two-term symmetrization is twice this value.

3.2 All mixed order derivatives

For every integer $q \geq 0$,

$$\partial_s^q w_n(s,a) = \frac{(-1)^q}{d!} \sum_{j=0}^{d} (-1)^j \binom{d}{j} \frac{\log^q(n+a+j)}{(n+a+j)^s}. \tag{3.5}$$

Differentiating (3.2) in any fixed collection of the independent s_ν therefore evaluates symmetrized sums with independent finite-difference logarithmic weights. Their right sides contain only finite coefficients times single derivatives $\zeta^{(j)}(s_\nu, a_\nu)$.

This is not a formal differentiation of integer-weight identities. The independent-variable identity is first established on a complex product domain with normal convergence. Subsequent integer specializations of its derivatives are consequently justified. Differentiating in the shifts as well produces mixed logarithmic and polygamma identities without changing that analytic foundation.

3.3 A useful distinction from general determinant sums

A generic determinant of zeta-valued moments contains products of zeta functions. Here their absence has an exact reason: the coefficient matrices of all the zeta coordinates share a one-dimensional image. No arithmetic cancellation conjecture is being used. Conversely, the formula does not say that an arbitrary unsymmetrized assignment of distinct orders has the same reduction. The symmetrization is essential to this particular determinant argument.

4 Gamma-weighted harmonic identities and differential ladders

4.1 The complete-harmonic weight at every integer order

For $q \geq 1$ define the local harmonic increment

$$h_q(n,a) = \sum_{j=0}^{d} \frac{1}{(n+a+j)^q}. \tag{4.1}$$

The complete exponential Bell polynomials are normalized by

$$\exp\left(\sum_{j\geq 1} x_j \frac{u^j}{j!}\right) = \sum_{k\geq 0} \mathcal{B}_k(x_1,\dots,x_k)\frac{u^k}{k!}, \qquad \mathcal{B}_0 = 1. \tag{4.2}$$

Theorem 4.1 (All-particle Gamma–harmonic evaluation)**.** *For every integer $m \geq 1$,*

$$w_n(m,a) = \frac{\Gamma(n+a)}{\Gamma(n+a+d+1)} \frac{\mathcal{B}_{m-1}(h_1, 1!h_2, \dots, (m-2)!h_{m-1})}{(m-1)!}. \tag{4.3}$$

For $m = 1$ the Bell factor is interpreted as one. For $m \geq 2$, the ordered r-particle sum of the products of these weights satisfies

$$\sum_{0\leq n_1<\cdots<n_r} \Delta(\boldsymbol{n})^2 \prod_{i=1}^{r} \left\{ \frac{\Gamma(n_i+a)}{\Gamma(n_i+a+d+1)} \frac{\mathcal{B}_{m-1}(h_1(n_i,a), \dots, (m-2)!h_{m-1}(n_i,a))}{(m-1)!} \right\}$$
$$= U_r(m,a) + \frac{(-1)^m}{(m-1)!} V_r(m,a)\psi^{(m-1)}(a). \tag{4.4}$$

At positive integral a, all the h_q are differences of ordinary generalized harmonic numbers. At rational a, they are rational-grid harmonic increments.

Proof. The elementary beta partial-fraction identity gives

$$w_n(1,a) = \frac{1}{(n+a)_{d+1}} = \frac{\Gamma(n+a)}{\Gamma(n+a+d+1)}.$$

One can also obtain it by integrating $t^{n+a-1}(1-t)^d/d!$ on $(0,1)$. Differentiate (2.1) $m-1$ times in a:

$$w_n(m,a) = \frac{(-1)^{m-1}}{(m-1)!}\partial_a^{m-1} w_n(1,a). \tag{4.5}$$

The logarithmic derivatives of the reciprocal rising factorial are

$$\partial_a^q \log w_n(1,a) = (-1)^q (q-1)! h_q(n,a).$$

The Bell formula for an exponential derivative, together with its weighted homogeneity, proves (4.3). Theorem 2.3 and the classical relation $\zeta(m,a) = (-1)^m \psi^{(m-1)}(a)/(m-1)!$ prove (4.4) [8]. □

The first weights are particularly transparent:

$$\begin{aligned}
w_n(2,a) &= \frac{h_1(n,a)}{(n+a)_{d+1}},\\
w_n(3,a) &= \frac{h_1(n,a)^2 + h_2(n,a)}{2(n+a)_{d+1}},\\
w_n(4,a) &= \frac{h_1(n,a)^3 + 3h_1(n,a)h_2(n,a) + 2h_3(n,a)}{6(n+a)_{d+1}}.
\end{aligned}$$

Thus the theorem evaluates nonlinear harmonic expressions, not only partial-fraction weights written in an unfamiliar notation.

For $r = 2$, set

$$L_n = \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3}, \qquad P_n = (n+1)(n+2)(n+3).$$

A concrete identity is

$$\boxed{\sum_{0 \le n < k} \frac{(k-n)^2 L_n L_k}{P_n P_k} = \frac{\pi^2 - 7}{16}.} \tag{4.6}$$

Replacing each L_n by $(L_n^2 + (n+1)^{-2} + (n+2)^{-2} + (n+3)^{-2})/2$ gives $(14\zeta(3) - 15)/32$. At three particles, the local interval has five terms and the same construction gives (2.15)–(2.16).

4.2 Spatial derivatives as exact insertion identities

For complex s and every integer $q \ge 0$,

$$\partial_a^q w_n(s,a) = (-1)^q (s)_q w_n(s+q,a). \tag{4.7}$$

Consequently, on the original convergence domain,

$$\partial_a^q \mathcal{Z}_r(s,a) = (-1)^q \sum_{k_1+\cdots+k_r=q} \frac{q!}{k_1!\cdots k_r!} \prod_{i=1}^{r} (s)_{k_i} \times \sum_{n_1<\cdots<n_r} \Delta(\boldsymbol{n})^2 \prod_i w_{n_i}(s+k_i,a). \tag{4.8}$$

All the coefficients and sums are explicit. The right side of (2.11) differentiates to

$$\partial_a^q \mathcal{Z}_r(s,a) = \partial_a^q U_r(s,a) + \sum_{j=0}^{q} \binom{q}{j} \partial_a^{q-j} V_r(s,a)(-1)^j (s)_j \zeta(s+j,a). \tag{4.9}$$

This uses the standard Hurwitz parameter-derivative law, so again only single zeta coordinates occur.

For example, let

$$\mathcal{I}_r(s,a) = \sum_{n_1<\cdots<n_r} \Delta(\boldsymbol{n})^2 \sum_{i=1}^{r} w_{n_i}(s+1,a) \prod_{j\neq i} w_{n_j}(s,a).$$

Then

$$\mathcal{I}_r(s,a) = V_r(s,a)\zeta(s+1,a) - \frac{\partial_a U_r(s,a) + \partial_a V_r(s,a)\zeta(s,a)}{s}. \tag{4.10}$$

For $\operatorname{Re} s > 1$, an anchored primitive is therefore

$$\int_{a_0}^{a} \mathcal{I}_r(s,x)\,\mathrm{d}x = -\frac{\mathcal{Z}_r(s,a) - \mathcal{Z}_r(s,a_0)}{s}, \qquad \operatorname{Re} a_0, \operatorname{Re} a > 0, \tag{4.11}$$

with the path lying in the right half-plane. The integral is path independent because the integrand is holomorphic there. Higher insertion combinations in (4.8) have the analogous anchored primitives obtained by reducing one derivative order. Thus the harmonic identities come with differential and antiderivative relations, not just isolated evaluations.

4.3 Rational shifts and arithmetic coordinates

For integer m, $U_r(m,a)$ and $V_r(m,a)$ are rational functions of a. At $a \in \mathbb{Q}_{>0}$, the all-particle value therefore lies in the $\mathbb{Q}$-linear span of 1 and $\zeta(m,a)$. For instance, the half-shift identity $\zeta(s,\frac{1}{2}) = (2^s-1)\zeta(s)$ gives ordinary-zeta reductions of (2.11) at $a = 1/2$. General rational shifts may instead be written with residue-class Dirichlet series or finite root filters. No claim of arithmetic independence of those coordinates is implied.

For noninteger s, the finite exponential factors in U_r, V_r remain well defined. Their continuation should not be confused with a statement that the corresponding *cutoff generating function* is always a finite combination of classical polylogarithms. The latter assertion is proved below only for rational weights, in particular at integer spectral orders.

5 A normal subtraction and all generalized Stieltjes jets

5.1 The correct escape amplitude

Write

$$\mathcal{Z}_{r,N}(s,a) = \sum_{0\le n_1<\cdots<n_r\le N} \Delta(\boldsymbol{n})^2 \prod_i w_{n_i}(s,a), \qquad \mathcal{Z}_{r,-1} = 0,$$

and define the largest-index block

$$B_N(s,a) = \mathcal{Z}_{r,N}(s,a) - \mathcal{Z}_{r,N-1}(s,a). \tag{5.1}$$

It vanishes for $N < r-1$. For $N \geq 1$, finite Cauchy–Binet on the remaining particles gives

$$B_N(s,a) = w_N(s,a) \det\left(\sum_{n=0}^{N-1} n^{i+j}(N-n)^2 w_n(s,a)\right)_{0\leq i,j<r-1}. \tag{5.2}$$

The amplitude that must be subtracted is

$$A_r(s,a) = c_d(s) V_r(s,a), \qquad c_d(s) = \frac{(s)_d}{d!}, \tag{5.3}$$

not just $V_r(s,a)$ away from $s=1$.

Lemma 5.1 (Uniform block remainder)**.** *Locally uniformly for* $\operatorname{Re} s > 0$, $\operatorname{Re} a > 0$,

$$B_N(s,a) = A_r(s,a)(N+a)^{-s} + O\big((N+1)^{-\operatorname{Re} s - 1}\big). \tag{5.4}$$

In particular the difference and all its fixed parameter derivatives are locally normally summable in that domain.

Proof. The $(r-1)$-particle ensemble with the same weight has total mass $V_r(s,a)$ for $\operatorname{Re} s > -1$: its largest required moment has degree $d-2$, and (2.7) without the zeta term extends to that domain by the identity theorem. For the present proof it suffices to work on a compact set with $\operatorname{Re} s \geq \sigma > 0$.

The absolute ensemble with one insertion of $\sum_i n_i$ is uniformly summable. Indeed, bounding its Vandermonde as in Lemma 2.1, the largest exponent after the insertion is at most $-\sigma - 1$. For $0 \leq n_i < N$,

$$0 \leq 1 - \prod_{i=1}^{r-1}(1 - n_i/N)^2 \leq \frac{2}{N}\sum_{i=1}^{r-1} n_i.$$

The sum of the absolute ensemble weights with some $n_i \geq N$ is also at most N^{-1} times the same first-moment majorant. Therefore

$$N^{-d}\frac{B_N(s,a)}{w_N(s,a)} = V_r(s,a) + O(N^{-1}), \tag{5.5}$$

where the quotient notation means the lower-particle polynomial sum, so no division by a zero weight is needed. This proof uses absolute majorants and is valid for complex s, a as well.

Combine (5.5) with (2.2) and $N^d(N+a)^{-d} = 1 + O(N^{-1})$ to obtain (5.4). Every bracket is holomorphic. Cauchy estimates on a slightly larger compact neighborhood give summable majorants for each fixed derivative; equivalently one can retain the resulting powers of $\log(N+1)$ in the estimate. □

Theorem 5.2 (Ordinary pole-cancelled series)**.** *The bracketed series*

$$T_r(s,a) = \sum_{N\geq 0}\big\{B_N(s,a) - A_r(s,a)(N+a)^{-s}\big\} \tag{5.6}$$

is locally normally convergent for $\operatorname{Re} s > 0$, $\operatorname{Re} a > 0$, *and*

$$\boxed{T_r(s,a) = U_r(s,a) + (1 - c_d(s))V_r(s,a)\zeta(s,a).} \tag{5.7}$$

The apparent pole at $s=1$ is removable. The right side gives an entire continuation in s; the assertion that the series itself is ordinary and normally convergent is limited to the stated half-plane.

Proof. Normal convergence is Lemma 5.1. For $\operatorname{Re} s > 1$, both unsubtracted series converge, so Theorem 2.3 gives (5.7) by subtraction. The identity theorem then gives the whole half-plane $\operatorname{Re} s > 0$. Since $c_d(1) = 1$, the unique pole of Hurwitz zeta cancels. The functions U_r, V_r, c_d are entire in s. □

5.2 An explicit triangular system for every Stieltjes constant

Let

$$U_k = \partial_s^k U_r(1,a), \quad A_k = \partial_s^k A_r(1,a), \quad b_k = \partial_s^k \{(1-c_d(s))V_r(s,a)\}\Big|_{s=1}. \tag{5.8}$$

Thus $b_0 = 0$. Also write $T_k = \partial_s^k T_r(1,a)$.

Theorem 5.3 (All-order Stieltjes identities)**.** *For every integer $k \geq 0$,*

$$T_k = \sum_{N\geq 0} \left\{ \partial_s^k B_N(1,a) - \frac{1}{N+a} \sum_{j=0}^{k} \binom{k}{j} A_j [-\log(N+a)]^{k-j} \right\} \tag{5.9}$$

$$= U_k + \frac{b_{k+1}}{k+1} + \sum_{j=1}^{k} \binom{k}{j} b_j (-1)^{k-j} \gamma_{k-j}(a). \tag{5.10}$$

The series is an ordinary locally normally convergent bracketed series. For $\operatorname{Re} a > 0$, it can be solved recursively for every $\gamma_m(a)$ because

$$b_1 = -H_d V_r(1,a) \neq 0. \tag{5.11}$$

Proof. Differentiate (5.6) under its normally convergent sum; Leibniz's rule yields (5.9). Set $s = 1+u$ in (5.7). Multiply

$$\sum_{j\geq 1} \frac{b_j}{j!} u^j \quad \text{by} \quad \frac{1}{u} + \sum_{m\geq 0} \frac{(-1)^m \gamma_m(a)}{m!} u^m.$$

The pole contributes $b_{k+1}/(k+1)$ to the kth derivative. The remaining product contributes the finite convolution in (5.10). Since $c_d'(1) = \sum_{j=1}^{d} 1/j = H_d$, equation (5.11) follows from Theorem 2.3. □

In explicit recursive form, putting $k = m+1$ gives

$$\gamma_m(a) = \frac{(-1)^m}{(m+1)b_1} \left\{ T_{m+1} - U_{m+1} - \frac{b_{m+2}}{m+2} - \sum_{j=2}^{m+1} \binom{m+1}{j} b_j (-1)^{m+1-j} \gamma_{m+1-j}(a) \right\}. \tag{5.12}$$

The empty sum at $m = 0$ is zero. Every coefficient in this triangular system is a finite combination of powers and logarithms at $a, a+1, \ldots, a+d-1$ and of ordinary harmonic numbers of length d.

The amplitude derivatives have a compact all-order description:

$$\log c_d(1+u) = \sum_{q\geq 1} \frac{(-1)^{q-1}}{q} H_d^{(q)} u^q, \qquad |u| < 1. \tag{5.13}$$

Consequently

$$c_d^{(k)}(1) = \mathcal{B}_k(H_d, -H_d^{(2)}, 2!H_d^{(3)}, \ldots, (-1)^{k-1}(k-1)!H_d^{(k)}).$$

Leibniz's rule with the finite determinants U_r, V_r now generates all the rows without a new infinite summation problem.

5.3 Concrete low-order identities

For $r = 2$, $a = 1$, put $L = \log 2$. The finite coefficient functions are

$$U(s) = -\frac{1}{2} + \frac{2^{-s}}{4}, \qquad V(s) = \frac{1 - 2^{-s}}{2}, \qquad c_2(s) = \frac{s(s+1)}{2}.$$

Equations (5.9)–(5.10) give

$$T_0 = -\frac{3}{4}, \tag{5.14}$$

$$T_1 = -\frac{1}{8} - \frac{1}{2}L - \frac{3}{8}\gamma, \tag{5.15}$$

$$T_2 = \frac{1}{2}L^2 - \frac{1}{4}L - \frac{1+3L}{4}\gamma + \frac{3}{4}\gamma_1. \tag{5.16}$$

For example, $A_0 = 1/4$ and $A_1 = 3/8 + L/4$, so the second line means precisely

$$\sum_{N\geq 0} \left\{ B'_N(1,1) - \frac{3/8 + L/4 - \frac{1}{4}\log(N+1)}{N+1} \right\} = -\frac{1}{8} - \frac{1}{2}L - \frac{3}{8}\gamma. \tag{5.17}$$

Here B'_N is computed by differentiating the finite two-particle block, using (3.5). The cancellation belongs to each displayed bracket; the two component series must not be split at $s = 1$.

5.4 Hard cutoffs are not spectral finite parts

Corollary 5.4 (The cutoff constant and the amplitude derivative)**.** *For $a > 0$, put $U_0 = U_r(1,a)$ and $V_0 = V_r(1,a)$. Then*

$$\lim_{N\to\infty} \{\mathcal{Z}_{r,N}(1,a) - V_0 \log(N+a)\} = U_0 + V_0\{\gamma_0(a) - H_d\}. \tag{5.18}$$

The spectral Laurent constant is

$$\mathrm{FP}_{s=1}\, \mathcal{Z}_r(s,a) = U_0 + \partial_s V_r(1,a) + V_0\gamma_0(a). \tag{5.19}$$

The spectral constant minus the hard-cutoff constant equals

$$\partial_s V_r(1,a) + H_d V_0 = \partial_s A_r(1,a). \tag{5.20}$$

Proof. The finite sum of the brackets in (5.6) at $s = 1$ is

$$\mathcal{Z}_{r,N}(1,a) - V_0 \sum_{n=0}^{N} (n+a)^{-1}.$$

It tends to $T_0 = U_0 - H_d V_0$. The digamma shift identity gives

$$\sum_{n=0}^{N} (n+a)^{-1} = \psi(N+a+1) - \psi(a),$$

and $\psi(N+a+1) - \log(N+a) \to 0$. This proves (5.18). Expanding (2.11) at $s = 1$ gives (5.19), and subtraction proves (5.20). □

For $r = 2, a = 1$, the constants are respectively

$$\frac{\gamma}{4} - \frac{3}{4}, \qquad \frac{\gamma}{4} - \frac{3}{8} + \frac{\log 2}{4}. \tag{5.21}$$

Their difference is $3/8 + \log 2/4$. The earlier determinantal report already warns that residue-amplitude derivatives affect this comparison [2]; the present calculation implements that warning for the new finite-difference spectral deformation. It is not a claim of an error in that source.

5.5 Stieltjes primitives with fixed normalization

For any $m \geq 0$ and $\operatorname{Re} a, \operatorname{Re} a_0 > 0$,

$$\int_{a_0}^{a} \gamma_m(x)\,\mathrm{d}x = \frac{(-1)^{m+1}}{m+1}\left\{\zeta^{(m+1)}(0,a) - \zeta^{(m+1)}(0,a_0)\right\}. \tag{5.22}$$

Indeed, comparing coefficients at $s = 0$ in $\partial_a \zeta(s,a) = -s\zeta(s+1,a)$ proves the derivative of (5.22); the anchor fixes its constant. This classical Hurwitz-jet primitive is consistent with the canonical manuscript's conventions [4, 8]. For $m = 0$, Lerch's identity reduces it to

$$\int_{a_0}^{a} \gamma_0(x)\,\mathrm{d}x = -\log\Gamma(a) + \log\Gamma(a_0).$$

Substituting (5.12) supplies ordinary determinant-series representatives for the integrands of these primitives. Local normal convergence permits termwise integration along every compact path in the right half-plane. Thus the Stieltjes series, Gamma primitive, and spectral derivative descriptions have one consistent normalization.

6 Classification of rank-one harmonic matrix pencils

6.1 The rational-weight model, including several lattices

Let $K \subseteq \mathbb{C}$ be a characteristic-zero field containing the parameters below. Consider a proper rational function

$$R(x) = \sum_{\nu=1}^{L} \sum_{j=0}^{J_\nu} \sum_{h=1}^{m_\nu} \frac{c_{\nu,j,h}}{(x + a_\nu + j)^h}, \qquad \operatorname{Re} a_\nu > 0. \tag{6.1}$$

Representatives from distinct lattices are chosen inequivalent modulo $\mathbb{Z}$, so that the associated harmonic families are not artificially duplicated. Zero coefficients and gaps in a lattice are allowed. For $q \geq 1$ put

$$H_N^{(q)}(a_\nu) = \sum_{n=0}^{N} (n + a_\nu)^{-q}. \tag{6.2}$$

The finite moment matrix is

$$M_N = \left(\sum_{n=0}^{N} n^{i+j} R(n)\right)_{0 \leq i,j < r}. \tag{6.3}$$

Proposition 6.1 (Explicit harmonic coefficient matrices)**.** *There is a matrix D_N with entries rational in N over K such that*

$$M_N = D_N + \sum_{\nu=1}^{L} \sum_{q=1}^{m_\nu} H_N^{(q)}(a_\nu)\, C_{\nu,q}. \tag{6.4}$$

Each $C_{\nu,q}$ is a fixed symmetric Hankel matrix. Writing $(C_{\nu,q})_{ij} = c_{\nu,q}[i+j]$, its entries are

$$c_{\nu,q}[k] = \sum_{h=q}^{m_\nu} \binom{k}{h-q} \sum_{j=0}^{J_\nu} c_{\nu,j,h}(-a_\nu - j)^{k-h+q}, \quad 0 \le k \le d. \tag{6.5}$$

A summand is omitted unless $0 \le h - q \le k$.

Proof. Expand $x^k = ((x + a_\nu + j) - (a_\nu + j))^k$. The coefficient of $(x + a_\nu + j)^{-q}$ in $x^k R(x)$ is exactly the corresponding summand in (6.5). Summing over $n = 0, \dots, N$ and shifting the finite harmonic sum changes it by

$$-\sum_{\ell=0}^{j-1} (\ell + a_\nu)^{-q} + \sum_{\ell=N+1}^{N+j} (\ell + a_\nu)^{-q},$$

which is rational in N over K. The polynomial part of $x^k R(x)$ sums to a polynomial in N, for instance by the finite Bernoulli-polynomial summation formula. All remaining terms therefore belong to D_N. □

Definition 6.2. The *harmonic matrix pencil* associated with R and r is

$$\mathscr{C}(\boldsymbol{X}) = \sum_{\nu,q} X_{\nu,q} C_{\nu,q}, \tag{6.6}$$

where the variables $X_{\nu,q}$ are algebraically independent. Its rank is the generic matrix rank over $K(\boldsymbol{X})$.

This definition is intrinsic after the distinct shift-orbits have been fixed. It does not posit algebraic independence of the actual functions $H_N^{(q)}(a_\nu)$ or of their limits.

6.2 The complete rank-one test

For $\tau \in K$, write

$$v(\tau) = (1, \tau, \dots, \tau^{r-1})^{\mathsf{T}}, \qquad \boldsymbol{e} = (0, \dots, 0, 1)^{\mathsf{T}}.$$

Theorem 6.3 (All rank-one harmonic pencils)**.** *The nonzero pencil (6.6) has rank one if and only if exactly one of the following forms is possible:*

$$C_{\nu,q} = \lambda_{\nu,q} v(\tau) v(\tau)^{\mathsf{T}} \qquad \text{for a common finite } \tau, \tag{6.7}$$

$$C_{\nu,q} = \lambda_{\nu,q} \boldsymbol{e}\boldsymbol{e}^{\mathsf{T}} \qquad \text{for all } \nu, q. \tag{6.8}$$

In either case some $\lambda_{\nu,q}$ is nonzero. Equivalently, the finite residue sums (6.5) obey either

$$c_{\nu,q}[k] = \lambda_{\nu,q} \tau^k \quad (0 \le k \le d)$$

with one common τ, or

$$c_{\nu,q}[k] = \lambda_{\nu,q} \mathbf{1}_{k=d} \quad (0 \le k \le d).$$

These are necessary and sufficient algebraic tests for the original rational weight, including repeated poles and multiple lattices.

Proof. If a symmetric pencil has rank at most one, every coefficient matrix has rank at most one. Two nonzero symmetric rank-one matrices can be written αuu^{T} and βvv^{T} over a field extension. If u, v are not proportional, some pair of coordinates has $u_i v_j - u_j v_i \neq 0$. The corresponding principal minor of $\alpha uu^{\mathsf{T}} + t\beta vv^{\mathsf{T}}$ is

$$t\alpha\beta(u_i v_j - u_j v_i)^2 \neq 0.$$

This contradicts rank at most one. All coefficient matrices consequently have one common rank-one direction.

It remains to classify a nonzero rank-one symmetric Hankel matrix. Write it as λuu^{T}. If $u_0 \neq 0$, normalize $u_0 = 1$. Equality of the entries $(0, j+1)$ and $(1, j)$ gives $u_{j+1} = u_1 u_j$, hence $u_j = \tau^j$. If $u_0 = 0$, let k be the first nonzero index. Were $k < r-1$, the Hankel entries (k, k) and $(k-1, k+1)$ would respectively be nonzero and zero. Thus $k = r-1$, giving the corner direction. The finite parameter τ belongs to K: it is the ratio of entries $(0,1)$ and $(0,0)$ of any coefficient matrix whose $(0,0)$ entry is nonzero. The converse is immediate from either displayed form. □

The test is constructive. If some $C_{\nu,q}(0,0) \neq 0$, compute $\tau = C_{\nu,q}(0,1)/C_{\nu,q}(0,0)$ and verify (6.7) in all entries. If no such coefficient exists, only (6.8) can hold. A zero pencil is reported separately, not mislabeled as rank one.

6.3 The critical-decay theorem: repeated finite differences

Let the residue polynomial for pole order q on lattice ν be

$$P_{\nu,q}(z) = \sum_{j=0}^{J_\nu} c_{\nu,j,q} z^j. \tag{6.9}$$

Lemma 6.4 (Corner support and finite-difference divisibility)**.** *The following statements are equivalent, with no critical-decay assumption:*

1. *Every $C_{\nu,q}$ is supported in its bottom-right entry.*
2. *For every ν, q and $0 \leq k < d$, $\sum_j c_{\nu,j,q}(-a_\nu - j)^k = 0$.*
3. *Every $P_{\nu,q}(z)$ is divisible by $(z-1)^d$ in $K[z]$.*
4. *There exists a unique proper rational function F on the same finite collection of translated lattices such that $R = \Delta^d F/d!$.*

The harmonic pencil is zero exactly when every $P_{\nu,q}$ is divisible by $(z-1)^{d+1}$, equivalently when R is a $(d+1)$st finite difference of a proper rational function.

Proof. Assume corner support. At the highest pole order $q = m_\nu$, (6.5) states directly that the moments of the residues of order q vanish through degree $d-1$. Descend in q. Every already treated term with $h > q$ involves a residue moment of degree $k - h + q < k < d$ and hence vanishes. The remaining term is exactly the required order-q moment. This proves the implication from the first to the second statement. Substitution in (6.5) proves the converse.

Shifting by a_ν does not change the span of polynomials of degree less than d. Ordinary powers of j and the falling factorials $j^{\underline{k}}$ span that same space. Thus the moment conditions are equivalent to $P_{\nu,q}^{(k)}(1) = 0$ for $k < d$, which is precisely the asserted divisibility.

For a proper rational function F with residue polynomials $Q_{\nu,q}$, replacing x by $x+1$ multiplies each residue polynomial by z. Therefore $\Delta^d F/d!$ has residue polynomials $(z-1)^d Q_{\nu,q}/d!$. Dividing $P_{\nu,q}$ explicitly constructs F and proves the fourth statement. Multiplication by $(z-1)^d$ is injective on polynomial residue data, so this proper rational F is unique. A polynomial kernel is excluded by properness.

For a zero pencil repeat the same descending argument through degree d, rather than $d-1$. This gives divisibility by $(z-1)^{d+1}$ and its stated finite-difference interpretation. □

Theorem 6.5 (Critical rational weights: an exact if-and-only-if)**.** *Suppose R has the form (6.1) and*

$$R(x) = \kappa x^{-(d+1)} + O(x^{-(d+2)}), \qquad \kappa \neq 0, \qquad d = 2r-2. \tag{6.10}$$

Then its harmonic pencil has rank one if and only if

$$\boxed{R(x) = \frac{1}{(2r-2)!}\Delta^{2r-2}F(x)} \tag{6.11}$$

for a proper rational function F on the same translated lattices. Equivalently, every residue polynomial (6.9) is divisible by $(z-1)^{2r-2}$. In that event every harmonic coefficient lies in the single bottom-right entry of the moment matrix.

Proof. The coefficient of x^{-1} in the expansion of $x^k R(x)$ at infinity is the sum of the coefficients of all its simple partial-fraction terms. By (6.10), it is zero for $k < d$ and κ for $k = d$. Proposition 6.1 therefore gives

$$\sum_{\nu=1}^{L} C_{\nu,1} = \kappa \boldsymbol{e}\boldsymbol{e}^{\mathsf{T}}. \tag{6.12}$$

If the harmonic pencil has rank one, the common-image argument in Theorem 6.3 forces all its coefficient matrices to be multiples of this nonzero corner matrix. Apply Lemma 6.4. Conversely, the finite-difference condition gives corner support. Equation (6.12) guarantees that the pencil is nonzero, hence has rank exactly one. □

This answers the source's repeated-pole rank-one question at its critical decay and supplies a complete finite residue test even away from that decay. The extension to several residue classes does not require a new determinant argument: equation (6.12) sums over those classes, while the finite-difference divisibility is checked separately on each shift-orbit.

If

$$F(x) = \sum_{\nu,j,q} \frac{b_{\nu,j,q}}{(x+a_\nu+j)^q}, \qquad R = \Delta^d F/d!,$$

then the corner coefficient is especially simple:

$$(C_{\nu,q})_{r-1,r-1} = \sum_j b_{\nu,j,q}. \tag{6.13}$$

Indeed, the dth residue moment of a normalized dth difference returns the leading coefficient of that polynomial moment; terms of higher pole order involve lower residue moments and vanish. For $F(x) = (x+a)^{-m}$ this coefficient is one at order m and zero at every other order. This recovers exactly the moment cancellation of Section 2 at integer spectral orders.

6.4 Why the nonzero critical coefficient matters

For $r = 2$, consider

$$R(x) = \frac{2}{(x+1)^2(x+2)(x+3)} = \frac{1}{(x+1)^2} - \frac{3}{2(x+1)} + \frac{2}{x+2} - \frac{1}{2(x+3)}. \tag{6.14}$$

It decays as $2x^{-4}$, faster than the critical x^{-3}. Its harmonic coefficient matrices are

$$C_1 = 0, \qquad C_2 = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = v(-1)v(-1)^{\mathsf{T}}.$$

The harmonic pencil has rank one, but $P_2(z) = 1$ is not divisible by $(z-1)^2$. Thus R is not a second difference of a proper rational function. This is the geometric branch in Theorem 6.3, not a counterexample to Theorem 6.5.

The example is supplied to prevent an overextension of the present result. It is not attributed to a false theorem in the inspected repository. In particular, one must not replace (6.10) by an unspecified "at least critical decay" in the if-and-only-if statement.

6.5 An exponential-jet version of the test

The residue tests can be compressed into finite exponential jets:

$$\mathcal{M}_{\nu,q}(t) = e^{-a_\nu t} \sum_{h=q}^{m_\nu} \frac{t^{h-q}}{(h-q)!} P_{\nu,h}(e^{-t}) \pmod{t^{d+1}}. \tag{6.15}$$

Expanding the right side shows that

$$\mathcal{M}_{\nu,q}(t) = \sum_{k=0}^{d} c_{\nu,q}[k] \frac{t^k}{k!}.$$

Thus rank one means either $\mathcal{M}_{\nu,q}(t) = \lambda_{\nu,q} e^{\tau t} \pmod{t^{d+1}}$ for one common τ, or $\mathcal{M}_{\nu,q}(t) = \lambda_{\nu,q} t^d/d! \pmod{t^{d+1}}$. This formulation gives a short certificate representation without constructing every matrix entry separately.

7 Rank-controlled polylogarithmic generators

7.1 What the harmonic rank does and does not imply

For the rational weight of Section 6, define

$$S_{r,N}(R) = \sum_{0 \le n_1 < \cdots < n_r \le N} \Delta(\boldsymbol{n})^2 \prod_i R(n_i) = \det M_N.$$

No infinite-sum convergence is needed for this finite identity. By Proposition 6.1, $S_{r,N}$ is a polynomial in shifted harmonic symbols with rational functions of N as coefficients.

Proposition 7.1 (Rank bounds harmonic degree)**.** *If the pencil $\mathscr{C}(\boldsymbol{X})$ has generic rank ρ, then*

$$\deg_{\boldsymbol{X}} \det(D_N + \mathscr{C}(\boldsymbol{X})) \le \rho. \tag{7.1}$$

In particular, a rank-one pencil makes $S_{r,N}$ affine in all the harmonic symbols, even with repeated poles and arbitrarily many particles. A zero pencil makes $S_{r,N}$ rational in N.

Proof. Over $K(N, \boldsymbol{X})$ a rank-ρ matrix is a product of an $r \times \rho$ and a $\rho \times r$ matrix. The determinant $\det(D_N + t\mathscr{C}(\boldsymbol{X}))$ therefore has degree at most ρ in t, by multilinearity or by the vanishing of minors of order larger than ρ. The coefficient of t^k is exactly the homogeneous degree-k part in $\boldsymbol{X}$. This proves the assertion without any assumption that D_N is invertible. □

The degree can be strictly smaller than ρ because of additional cancellations involving D_N. Neither ρ nor the polynomial degree proves minimal depth of evaluated periods. These are finite structural upper bounds.

7.2 A constructive depth bound on rational grids

For positive integers $p_1, \dots, p_\ell$, our multiple-polylogarithm convention is

$$\mathrm{Li}_{p_1,\dots,p_\ell}(z_1, \dots, z_\ell) = \sum_{n_1 > \cdots > n_\ell \ge 1} \frac{z_1^{n_1} \cdots z_\ell^{n_\ell}}{n_1^{p_1} \cdots n_\ell^{p_\ell}}. \tag{7.2}$$

Its depth is ℓ. In the disk, all identities below use the common power-series germ; rational Laurent prefactors have removable singularities only after the full expression is assembled.

Theorem 7.2 (Harmonic rank controls cutoff-generator depth)**.** *Assume every a_ν is positive rational, and choose a common denominator D for the pole locations. If the harmonic pencil has rank ρ, then*

$$G_R(z) = \sum_{N \ge 0} S_{r,N}(R) z^{N+1}, \qquad |z| < 1, \tag{7.3}$$

can be expressed by finitely many cyclotomic multiple polylogarithms of level dividing D and depth at most $\rho + 1$, rational functions of $z^{1/D}$, and Euler derivatives of such expressions. Every positive Euler primitive, normalized to vanish at $z = 0$, obeys the same depth bound.

For integer-grid poles, no fractional power or root-of-unity filter is needed. In particular, the rank-one families in Theorems 6.3–6.5 have depth at most two. A zero pencil has depth at most one.

Proof. Proposition 7.1 reduces each coefficient to a finite sum of terms

$$Q(N) \prod_{j=1}^{\ell} H_N^{(q_j)}(a_{\nu_j}), \qquad \ell \le \rho, \tag{7.4}$$

where Q is rational and has rational-grid poles. All such sequences have at most polynomial-logarithmic growth, so their generating series converge normally in the disk.

First consider integral shifts. Expand the product of finite harmonic sums by partitioning the index set according to equality and strict ordering. This is the finite quasi-shuffle argument; it yields nested sums of depth at most ℓ. At an equal index, different shifted rational factors are reduced by ordinary partial fractions. Integer shifts of a cutoff can be removed by subtracting a finite boundary

strip. Removing one such strip either changes a rational coefficient or reduces the number of nested free indices; it cannot increase depth.

Next decompose $Q(N)$ into a polynomial and partial fractions. For an unshifted denominator, the outer summation adds at most one strict index to the nested harmonic sum. Its diagonal contribution has smaller depth. For example,

$$\sum_{M\geq 1} \frac{z^M H_M^{(q)}}{M^p} = \mathrm{Li}_{p,q}(z,1) + \mathrm{Li}_{p+q}(z), \tag{7.5}$$

$$\sum_{M\geq 1} z^M H_M^{(q)} = \frac{\mathrm{Li}_q(z)}{1-z}. \tag{7.6}$$

Shifting an outer denominator replaces M by $M+j$, followed by the same finite boundary-strip procedure. Polynomial factors in M are implemented by the Euler operator $z\partial_z$. These operations prove the bound $\ell+1$ on an integer grid.

For rational shifts write $a = A/D$ after moving the representative into $0 < a \leq 1$ and accounting for finitely many initial terms. Then

$$H_N^{(q)}(A/D) = D^q \sum_{\substack{1\leq m\leq DN+A \\ m\equiv A \ (D)}} m^{-q}.$$

Resolve a congruence by the exact filter

$$\mathbf{1}_{m\equiv A\ (D)} = \frac{1}{D}\sum_{j=0}^{D-1} \omega^{j(m-A)}, \qquad \omega = e^{2\pi i/D}.$$

After rescaling indices, strict inequalities between the original integer indices differ from strict inequalities between the new residue-class indices only on equal-block diagonals. Those diagonals are reduced by partial fractions and have fewer free indices. Induction on their number reduces to strictly nested sums of the form (7.2), with roots of unity in the arguments and $z^{1/D}$ in the outer argument. Depth does not increase. This also explains why different residue classes can be handled together.

Finally, a positive Euler primitive divides the coefficient of z^{N+1} by a positive power of $N+1$. It changes only the rational factor $Q(N)$ in (7.4), leaving the maximum number of harmonic factors unchanged. The same reduction applies. □

This is a representation theorem, not an irreducibility theorem. It also does not assert that every endpoint reduces to ordinary zeta values and their products. Repeated poles introduce inner indices larger than one, for which the endpoint reduction problem is different from the simple-pole family in the earlier report.

7.3 An explicit repeated-pole $\mathrm{Li}_{2,2}$ identity

Take $r=2$, $a=1$, $s=2$, and write

$$S_N = \mathcal{Z}_{2,N}(2,1), \qquad M = N+1.$$

The weight has repeated poles,

$$w_n(2,1) = \frac{1}{2}\left\{\frac{1}{(n+1)^2} - \frac{2}{(n+2)^2} + \frac{1}{(n+3)^2}\right\}.$$

A finite calculation from (2.8) gives

$$S_{M-1} = P(M) + Q(M)H_M^{(2)}, \qquad M \geq 1, \tag{7.7}$$

where

$$P(M) = -\frac{7}{16} - \frac{3}{4(M+2)} + \frac{7}{4(M+2)^2} + \frac{7}{8(M+1)^2} - \frac{1}{2(M+1)^4}, \tag{7.8}$$

$$Q(M) = \frac{3}{8} - \frac{1}{2(M+1)^2} + \frac{1}{2(M+2)^2}. \tag{7.9}$$

For example, $M = 1$ gives zero, as required because there are not yet two distinct indices.

Theorem 7.3 (Closed repeated-pole cutoff generator)**.** *For $|z| < 1$,*

$$\sum_{M\geq 1} S_{M-1} z^M = \frac{1-z}{2z^2}\operatorname{Li}_{2,2}(z,1) + \frac{10-7z}{8z^2(1-z)}\operatorname{Li}_2(z) + \frac{4z-7}{4z^2}\operatorname{Li}_1(z) - \frac{\operatorname{Li}_4(z)}{2z} + \frac{1}{2z} - \frac{7}{16(1-z)}. \tag{7.10}$$

The apparent singularity at $z = 0$ is removable, and the resulting function begins at order z^2.

Proof. For completeness, the three finite moments, with $H = H_M^{(2)}$, are

$$\mu_{0,M-1} = \frac{M(3M^3+18M^2+39M+28)}{8(M+1)^2(M+2)^2},$$

$$\mu_{1,M-1} = \frac{M(M-1)(M^2+7M+8)}{8(M+1)^2(M+2)^2},$$

$$\mu_{2,M-1} = H - \frac{9M^4+70M^3+133M^2+76M}{8(M+1)^2(M+2)^2}.$$

The determinant $\mu_0\mu_2 - \mu_1^2$ gives (7.7)–(7.9) by rational arithmetic.

The shifted harmonic sums needed to sum that expression are

$$\sum_{M\geq 1} \frac{z^M H_M^{(2)}}{(M+1)^2} = z^{-1}\operatorname{Li}_{2,2}(z,1),$$

$$\sum_{M\geq 1} \frac{z^M H_M^{(2)}}{(M+2)^2} = z^{-2}\left\{\operatorname{Li}_{2,2}(z,1) - (1+z)\operatorname{Li}_2(z) - 2(1-z)\operatorname{Li}_1(z) + 3z\right\}.$$

To verify the second line, put $L = M + 2$ and remove the missing term from $H_{L-1}^{(2)}$. The remaining rational factor is reduced using

$$\frac{1}{L^2(L-1)^2} = \frac{1}{L^2} + \frac{1}{(L-1)^2} + \frac{2}{L} - \frac{2}{L-1}.$$

The plain shifted sums are

$$\sum_{M\geq 1} \frac{z^M}{(M+j)^p} = z^{-j}\left\{\operatorname{Li}_p(z) - \sum_{\ell=1}^{j}\frac{z^\ell}{\ell^p}\right\}.$$

Insert these identities and (7.6) into (7.7). Collecting terms proves (7.10). All steps are absolutely convergent in the disk. The original series proves removability and its initial vanishing; no separate branch assignments at the apparent poles are made. □

As $z \uparrow 1$, Abel's theorem and the convergent sequence S_N give

$$\lim_{z\uparrow 1}(1-z)\sum_{M\geq 1} S_{M-1}z^M = \frac{3}{8}\zeta(2) - \frac{7}{16} = \frac{\pi^2-7}{16}. \tag{7.11}$$

This independently visible endpoint agrees with (4.6). The depth-two coordinate in the cutoff generator disappears from this particular Abel limit because its coefficient has an additional factor $1-z$. This is an exact cancellation, not a general theorem of depth-two endpoint reducibility.

7.4 Constructing new examples without a search

Choose any proper rational seed F with poles on positive rational translated grids and set $R = \Delta^{2r-2}F/(2r-2)!$. Lemma 6.4 proves that its cutoff partition has an affine harmonic expression and hence a depth-two generator at most. This works for seeds with several pole orders and several residue classes. If the seed is itself a first difference, the harmonic pencil vanishes and the generator has depth at most one.

For example, any seed of the form

$$F(x) = \frac{u}{x+1/3} + \frac{v}{(x+1/2)^2} + \frac{w}{(x+2/3)^3}$$

produces an all-particle rank-at-most-one family on level six. The corner entries are u, v, w in their respective harmonic coordinates by (6.13). When $u \neq 0$, the decay is nonzero critical and Theorem 6.5 applies. The theorem supplies a finite construction and a complete rank certificate; it does not require a numerical integer-relation search.

8 Source audit, corrections, and verification status

8.1 The source anchor and the inspection actually performed

The repository was anchored at commit

`73f18ef013d5b74eafc883efb2c15210642df12a`,

whose timestamp is 11 October 2026 at 18:22:01 UTC. Both directories in the research request were inspected at that pin. The source record in `provenance/sources.json` lists the relevant paths and the level of inspection, rather than implying a full audit of the repository.

The targeted reading included the canonical main file and preface, the research-status chapter, the opening 280 lines of the integration chapter, and the adjacent-tail and research-question sections of the differential polygamma continuation. The determinantal harmonic-resonance report was read for its defining ensemble, Gauss product, first-resonance context, stated simple-pole harmonic-rank result, and further research questions. Its requested repeated-pole classification and rational-grid extension are the direct targets of Section 6.

The incoming directory contained six archives at the anchor. The member list and selected status/readme material of *Polygamma Reconstruction and Resonance* and the readme of *Harmonic Tails, Shifts and Twists* were inspected. A combined retrieval also exposed readme material from *Harmonic Tail Gamma Continuation*, but it was not treated as a full interior audit. The Beta Crossing, Quadratic Spectrum, and Same-Direction Complex Orders packages were not read in full

in this continuation. No error or absence of a theorem is attributed to their unread contents. This targeted comparison is enough to identify the specific source question being answered, but not to prove an exhaustive novelty claim.

8.2 Proposed status updates

The source question about rank-one harmonic pencils should now be split into a proved classification and remaining arithmetic-depth questions. Theorem 6.3 supplies the full residue test in the specified rational-weight model. Theorem 6.5 gives the particularly short critical-decay characterization, allowing repeated poles and multiple rational residue classes. The source's simple-pole construction is thereby extended, not invalidated.

The source question about harmonic rank controlling depth receives the constructive upper bound in Theorem 7.2. It should not be rewritten as a theorem about *minimal* depth. The possible strict inequality between harmonic rank, determinant degree, and arithmetic depth remains an important separate issue.

The next-two-particle-resonance question in the older report should be cross-linked to the current incoming reconstruction/resonance package, subject to the repository's usual integration review. The present finite-difference deformation should be listed separately: its simple pole and its coefficient $c_d(s)$ are not the spectral excess and pole structure of every higher Gauss resonance.

8.3 Corrections and safeguards established here

No newly false analytic theorem was found in the portions of the existing material inspected for this article. The following are precise integration safeguards, with explicit examples rather than allegations about unread sources.

First, the critical-decay characterization requires a *nonzero* coefficient κ in (6.10). The faster-decaying weight (6.14) has a rank-one geometric pencil and is not a dth finite difference. Replacing exact critical decay by merely a big-O condition would make the new theorem false.

Second, the escape amplitude is $A_r = c_d V_r$. A counterterm using V_r for all s gives the wrong derivative rows. In particular, the pole-times- next-coefficient term $b_{k+1}/(k+1)$ in (5.10) must be retained.

Third, the hard-cutoff constant and the spectral Laurent constant differ by $\partial_s A_r(1, a)$. Equation (5.21) gives a nonzero elementary discrepancy. The earlier source already recognizes this distinction; our formulas make it explicit for this family.

Fourth, meromorphic continuation does not commute with an isolated pointwise specialization of a nonnormally convergent family. At $s = 0$ the weights vanish, but the continued value is $-1/4$ for two particles. This also limits the interpretation of identities at nonpositive orders.

Finally, finite harmonic rank and polylogarithmic representation depth are not arithmetic independence. An explicit $\mathrm{Li}_{2,2}$ term is not proof that depth two is necessary. Conversely, the simple-pole report's special ordinary-zeta endpoint closure should not be extrapolated to all repeated-pole cases without a separate reduction argument.

8.4 What the supplied verification proves

The exact suite `verification/verify_exact.py` performs **1,037 assertions** in rational or symbolic arithmetic. The machine-readable record is `results/exact_checks.json`. It includes 576 finite

endpoint moment identities, 18 direct ordered-sum Cauchy–Binet comparisons, 18 residue-product checks, 12 single-zeta matrix identities, independent-order/shift polarization tests, 144 Gamma–harmonic Bell identities, finite-difference residue inversions, zero-pencil checks, the geometric counterexample, and 20 deliberately perturbed weights rejected by the rank-one test. It also checks the explicit cutoff formula, 44 generating-function coefficients, and the Stieltjes/Bell coefficient rules.

The independent numerical suite records **29 diagnostics** in `results/numerical_checks.json`. Nine compare determinants of Mellin-integral moments with the finite-endpoint/zeta expression, including noninteger and complex orders and shifts. Two test mixed-order/shift Mellin identities. Four compare direct finite sums with finite boundary formulas. Four test hard-cutoff constants through finite formulas at $N = 10^{20}$; six test differentiated Stieltjes brackets at $N = 10^{18}$. Three test the explicit polylogarithmic generator, and one tests the anchored Gamma primitive.

The standard working precision was 60 decimal digits, with increased precision for cutoff cancellations. The Mellin-determinant relative residuals in the recorded run are below 6.1×10^{-57}. The largest relative discrepancy across *all* diagnostics is about 8.83×10^{-13}, occurring in a high derivative of a finite-cutoff approximation. It would therefore be incorrect to describe the whole suite as agreement to 60 digits. Each row records its actual discrepancy, tolerance, and cutoff where applicable. These are floating-point diagnostics, not certified interval enclosures.

The infinite-family conclusions rest on the proofs in this article. Finite exact checks test implementation and normalization; they do not replace the general algebraic or analytic arguments. A successful PDF build likewise verifies document compilation, not mathematical truth.

8.5 Integration boundary

The package is self-contained and includes standalone and modular TeX, the compiled PDF, verification programs and result records, a build script, a source-scope record, an explicit claim-status file, and checksum verification. The suggested repository destination is recorded in `integration/INTEGRATION.md`. No repository files were changed, no archive was retired, and no commit or pull request was created. The manuscript's Gaussian S_6 and revised S_8 proof statuses are unchanged [5].

9 Further research questions

9.1 Higher-rank harmonic pencils and their moduli

The rank-one case has the two exact forms (6.7)–(6.8). Classify the rational residue arrays for which the generic harmonic pencil has rank at most two, then at most ρ. Hankel structure suggests a connection with finite exponential-polynomial moment sequences and confluent Vandermonde decompositions. The difficulty is the requirement that *all* coefficient matrices lie in one low-rank linear space, not merely that each matrix separately has low rank. An effective necessary and sufficient classification would extend the present finite residue certificate beyond rank one.

9.2 When does determinant degree equal harmonic rank?

For a given rational weight, let δ be the total harmonic degree of $\det(D_N + \mathscr{C}(\boldsymbol{X}))$. We proved $\delta \le \rho$. Determine exact conditions for strict inequality, retaining the dependence of D_N on the same residue data as the pencil. A useful target is a criterion in minors and discrete residues that avoids

expanding the full determinant. This would sharpen the formal polylogarithmic depth bound from $\rho + 1$ to $\delta + 1$ before any functional reductions are used.

9.3 Minimal depth in a specified functional relation system

The generator (7.10) contains $\mathrm{Li}_{2,2}(z, 1)$, but its coefficient could interact with functional equations in a broader alphabet. Determine whether this generator reduces to classical polylogarithms in a precisely declared differential or symbol calculus. A nonzero obstruction would concern that formal system, not numerical period independence. The problem is especially concrete because all rational coefficients are explicit and the endpoint cancellation is already known.

9.4 Closed all-rank formulas for the finite exponential coefficients

The determinants U_r, V_r are finite at every rank, and $V_r(1, a)$ has the Gamma product (2.12). Find short recurrences or product formulas for $U_r(m, a)$ and $V_r(m, a)$ at fixed integer $m \geq 2$, or for their jets at $s = 1$. The residue-product proof suggests using finite discrete orthogonal polynomials at $m = 1$ followed by structured confluent perturbations. A successful result should improve symbolic expression growth without introducing hidden infinite coordinates.

9.5 Unsymmetrized mixed-order sums

The polarized determinant evaluates the symmetrized sum with independent orders and shifts. What additional coordinates are required for an individual ordered assignment? For two particles, the sum and difference under exchanging the two labels are natural first targets. The sum is already evaluated by (3.3); the antisymmetric component may connect to multiple Hurwitz zeta or Lerch functions. One should seek an identity on a genuine complex-parameter domain before differentiating at integer values.

9.6 Noninteger-order cutoff generators

The spectral identity holds for arbitrary complex orders, but the rational partial-fraction proof of polylogarithmic cutoff closure uses integer orders. Determine a natural finite function class for the cutoff generator at noninteger s. A likely candidate involves multiple Hurwitz–Lerch functions with independently shifted denominators, but merely naming such a function is not a reduction. The useful target is a finite transform or differential system that specializes to Theorem 7.3 and preserves its initial conditions.

9.7 More general annihilating stencils

Replace the consecutive finite difference by $\sum_j \alpha_j f(x + \lambda_j)$, with distinct rational shifts and $\sum_j \alpha_j \lambda_j^k = 0$ for $k < d$. Determine which stencils still confine the moment special functions to one matrix direction. When the shifts are integers, residue-polynomial divisibility is the right language; on nonuniform grids, endpoint terms may produce several Lerch coordinates. A classification should distinguish a genuinely rank-one common image from a mere vanishing of the first asymptotic coefficients.

9.8 Higher Gauss resonances remain a distinct problem

For the earlier Gauss ensemble, the resonances $b - a - c = 2r - 2 - M$ can have several divergent moment entries and overlapping escape regimes. Our spectral deformation has only the single Hurwitz pole and therefore does not resolve that general problem. Extend the largest-index proof with a canonical subtraction of multiple escape regions, and compare its cutoff polynomial with the Laurent coefficients of the known Gamma product. The next two-particle case has a separate incoming treatment; the remaining all-rank statement should retain that provenance and its more complicated pole structure.

9.9 Endpoint Euler primitives and weight drops

Theorem 7.2 bounds the depth of every normalized Euler primitive inside the disk. Determine which absolutely convergent endpoint combinations of the integer-order finite-difference family reduce to ordinary zeta values or products, and which require genuine multiple zeta coordinates in a stated relation system. The cancellation in (7.11) is one exact example, not evidence for an unrestricted closure claim. A systematic endpoint formula would connect the present cutoff identities with the repository's harmonic-tail and cyclotomic reduction programs.

9.10 Formal verification with a small analytic interface

The finite residue matrices, divisibility tests, polynomial annihilation, and Cauchy–Binet normalization are suitable exact formalization targets. They can be separated from three analytic interfaces: local normal convergence of the spectral ensemble, the uniform largest-index remainder, and the Hurwitz Laurent expansion. A proof-assistant implementation of that separation would make subsequent repeated-pole identity generation substantially more reliable. The supplied scripts are executable certificates and diagnostics, not such a formalization.

A Derivation of the Gamma product for the residue

The following determinant is a classical finite algebraic ingredient of the source's Gauss–Vandermonde proof. We include its derivation so the normalization in (2.12) can be checked independently.

Lemma A.1 (A rational factorial determinant)**.** *For indeterminates $x_0, \dots, x_{L-1}, t$,*

$$\det_{0\le i,j<L} \frac{(x_i)_j}{(x_i+t)_j} = \frac{\displaystyle\prod_{j=0}^{L-1}(t)_j \prod_{0\le i<k<L}(x_k - x_i)}{\displaystyle\prod_{i=0}^{L-1}(x_i+t)_{L-1}}. \tag{A.1}$$

It is an identity of rational functions and hence holds wherever the specialized expressions are defined, with removable values interpreted by cancellation.

Proof. Multiply row i by $(x_i+t)_{L-1}$. Column j becomes the polynomial

$$p_j(x) = (x)_j(x+t+j)_{L-1-j},$$

of degree at most $L-1$. Its evaluation determinant is the Vandermonde in the x_i times the determinant of the polynomial coefficients. Evaluate at $x_i = -i$. The matrix is lower triangular, since $(-i)_j = 0$ for $j > i$, and its diagonal is $(-1)^i i!(t)_{L-1-i}$. Dividing its determinant by the Vandermonde at $0, -1, \ldots, -(L-1)$ leaves $\prod_{j=0}^{L-1}(t)_j$. Dividing out the original row factors proves the identity. □

Set $L = r-1$ and $d = 2r-2$. Since the two polynomial bases $n^{\underline{i}}$ and $(n+a)_j$ are monic in their respective degrees, Cauchy–Binet gives

$$V_r(1,a) = \det_{0\le i,j<L} \sum_{n\ge 0} n^{\underline{i}}(n+a)_j \frac{\Gamma(n+a)}{\Gamma(n+a+d+1)}. \tag{A.2}$$

Terms $n < i$ vanish. Shift $n = i+k$ and apply Gauss summation [10]. Because $i+j \le 2r-4 = d-2$, its excess $d-i-j$ is positive. The resulting entry is

$$M_{ij} = \frac{\Gamma(i+1)\Gamma(a+i+j)\Gamma(d-i-j)}{\Gamma(d+1-j)\Gamma(a+d)}. \tag{A.3}$$

Factor the row terms $\Gamma(i+1)\Gamma(a+i)\Gamma(d-i)$, the column terms $(-1)^j/\Gamma(d+1-j)$, and one factor $1/\Gamma(a+d)$ per row. The remaining determinant is

$$\det_{0\le i,j<L} \frac{(a+i)_j}{(1-d+i)_j}.$$

Apply Lemma A.1 with $x_i = a+i$ and $t = 1-d-a$. None of the denominator rising factorials vanishes in the required range of indices. Its Vandermonde is $\prod_{k=0}^{L-1} k!$. Substitution gives the intermediate product

$$\begin{aligned} V_r(1,a) = {} & \frac{\prod_{i=0}^{L-1}\Gamma(i+1)\Gamma(a+i)\Gamma(d-i)}{\Gamma(a+d)^L} \prod_{j=0}^{L-1} \frac{(-1)^j}{\Gamma(d+1-j)} \\ & \times \frac{\prod_{j=0}^{L-1}(1-d-a)_j \prod_{k=0}^{L-1} k!}{\prod_{i=0}^{L-1}(1-d+i)_{L-1}}. \end{aligned} \tag{A.4}$$

Using

$$(1-d-a)_j = (-1)^j \frac{\Gamma(a+d)}{\Gamma(a+d-j)}, \qquad (1-d+i)_{L-1} = (-1)^{L-1}\frac{\Gamma(d-i)}{\Gamma(d-i-L+1)},$$

and $d-L+1 = r$, the signs cancel and (A.4) becomes

$$\prod_{j=0}^{r-2} \frac{j!\,\Gamma(a+j)\Gamma(j+1)\Gamma(r-j)}{\Gamma(2r-1-j)\Gamma(a+2r-2-j)}.$$

This proves (2.12) for $\operatorname{Re} a > 0$. In particular,

$$V_2(1,a) = \frac{1}{2a(a+1)}, \qquad V_3(1,a) = \frac{1}{72a(a+1)^2(a+2)^2(a+3)}.$$

No additional $L!$ is introduced, since the defining ensemble uses strictly ordered indices throughout.

B Finite algorithms and replayable certificates

B.1 Computing the single-zeta answer

For given r, order s, and shift a, compute the $2r-1$ finite endpoint expressions (2.5). Form their $r \times r$ Hankel matrix and its upper-left $(r-1) \times (r-1)$ minor. Their determinants are U_r, V_r. The result is then (2.11). At integer $s = m$, all entries are rational functions of a; at rational a, exact rational arithmetic suffices to produce the two coefficients. The program `verification/core.py` implements this construction without evaluating any zeta function.

For a finite cutoff N, add the finite harmonic term and the upper boundary (2.9) before taking the determinant. This produces an exact formula whose number of endpoint terms depends on r, not on N. The large numerical cutoffs in the verification record were handled by this formula, not by enumerating 10^{20} summands.

B.2 Classifying a rational weight

A complete certificate consists of its partial-fraction residues, the matrices (6.5), and one of the following outputs: zero pencil; corner pencil; geometric pencil with the exact parameter τ; or a nonzero minor of a linear combination disproving rank one. At critical decay, the residue-polynomial quotients

$$Q_{\nu,q}(z) = d!\, P_{\nu,q}(z)/(z-1)^d$$

are particularly compact positive certificates. Their coefficients reconstruct the unique proper rational seed F.

The implementation checks rank-one support exactly; it does not replace formal harmonic variables by numerical evaluations. Its deliberately altered examples add a single pole of order $d+2$. This preserves the nonzero critical coefficient while breaking the divisibility condition, and the altered pencil is rejected in every recorded case.

B.3 Generating the Stieltjes rows

To obtain row k, only finite derivatives of U_r, V_r, c_d through orders k, $k+1$ are needed. Use (5.13) for c_d, ordinary finite differentiation for the endpoint powers, and the coefficient rule (5.10). The next coefficient b_{k+1} is compulsory because it multiplies the pole u^{-1}.

The exact suite tests this rule with independent formal symbols for the Stieltjes constants. This verifies the coefficient extraction separately from special-function numerics. Its analytic realization as an ordinary series is the distinct content of Lemma 5.1 and Theorem 5.2.

B.4 Numerical representations used for independent checks

For $0 \le k \le d$, the moment integral is

$$\mu_k(s,a) = \frac{1}{d!\Gamma(s)} \int_0^\infty t^{s-1} e^{-at} (1-e^{-t})^d \left(\sum_{n \ge 0} n^k e^{-nt} \right) \mathrm{d}t. \tag{B.1}$$

For $k = 0$, the series in parentheses is $(1-e^{-t})^{-1}$; for $k > 0$, it is a rational function obtained by repeated application of $q\partial_q$ to $(1-q)^{-1}$ at $q = e^{-t}$. Thus this evaluator uses only an elementary Mellin kernel and Gamma normalization, not the finite endpoint formula being tested.

The numerical program computes the determinant of these integral moments and compares it with $U_r + V_r\zeta$. Near $t = 0$, it evaluates $1 - e^{-t}$ through an exponential-minus-one routine to avoid cancellation. The hard-cutoff diagnostics instead use the finite endpoint formula with increased precision. Their errors include actual finite-cutoff remainders, so their reported accuracy is intentionally different from that of the Mellin comparisons.

B.5 Rebuilding and verifying the delivery

From the package directory, run

```
python verify_manifest.py
python verification/verify_exact.py
python verification/verify_numeric.py
python build.py
```

The two mathematical verification commands regenerate their JSON result records. The build script regenerates the standalone TeX and runs the LaTeX compiler. The checksum verifier checks the delivered files before modification; after regenerating timestamped result records or a PDF, expected byte hashes can differ. No network access is needed for either verification suite. The exact dependency versions and the TeX package requirements are recorded in the accompanying files.

References

[1] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*, canonical manuscript, main file and preface. Repository snapshot `73f18ef013d5b74eafc883efb2c15210642df12a`. https://github.com/VladimirReshetnikov/ProveIt/tree/73f18ef013d5b74eafc883efb2c15210642df12a/Analysis/Polylogarithms/docs/manuscript.

[2] ProveIt Contributors, *Determinantal Harmonic Resonance: Gauss–Vandermonde finite parts, Stieltjes jets, and depth-two polylogarithmic closure*, preserved report, `article_standalone.tex`, at the same snapshot. https://github.com/VladimirReshetnikov/ProveIt/tree/73f18ef013d5b74eafc883efb2c15210642df12a/Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/determinantal-harmonic-resonance. The repeated-pole, harmonic-rank, and rational-grid questions in its further-research section are the direct targets of this continuation.

[3] ProveIt Contributors, *Incoming research packages*, directory catalogue and selected status/readme material, inspected 11 October 2026 at `73f18ef013d5b74eafc883efb2c15210642df12a`. https://github.com/VladimirReshetnikov/ProveIt/tree/73f18ef013d5b74eafc883efb2c15210642df12a/docs/incoming. The precise limits of interior inspection are stated in Section 8.

[4] ProveIt Contributors, *Integration chapter*, canonical manuscript `chapters/07-integration.tex`, opening 280 source lines inspected at the same snapshot.

[5] ProveIt Contributors, *Discovery and proof-status chapter*, canonical manuscript `chapters/10-discovery.tex`, at the same snapshot.

[6] S. Chen and M. F. Singer, *Residues and Telescopers for Rational Functions*, arXiv:1203.4200 (2012). https://arxiv.org/abs/1203.4200.

[7] C. E. Arreche and H. P. Sitaula, *A computational approach to rational summability and its applications via discrete residues*, arXiv:2503.15636 (2025). https://arxiv.org/abs/2503.15636. Cited for the broader discrete-residue and rational-summability framework, not as a source of the determinant identities proved here.

[8] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*; in particular the defining series, shift and parameter-derivative formulas, polygamma values, and Lerch's identity at spectral order zero. https://dlmf.nist.gov/25.11.

[9] NIST Digital Library of Mathematical Functions, Chapter 5, *Gamma Function*, especially §5.2 and §5.15. https://dlmf.nist.gov/5.2; https://dlmf.nist.gov/5.15.

[10] NIST Digital Library of Mathematical Functions, §15.4, equation 15.4.20, Gauss hypergeometric summation. https://dlmf.nist.gov/15.4.E20.

[11] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*, and §25.14, *Lerch's Transcendent*. https://dlmf.nist.gov/25.12; https://dlmf.nist.gov/25.14. The multiple-polylogarithm convention is defined explicitly in (7.2).